Sai Charan
02/03/26

SOC Task 2 Report

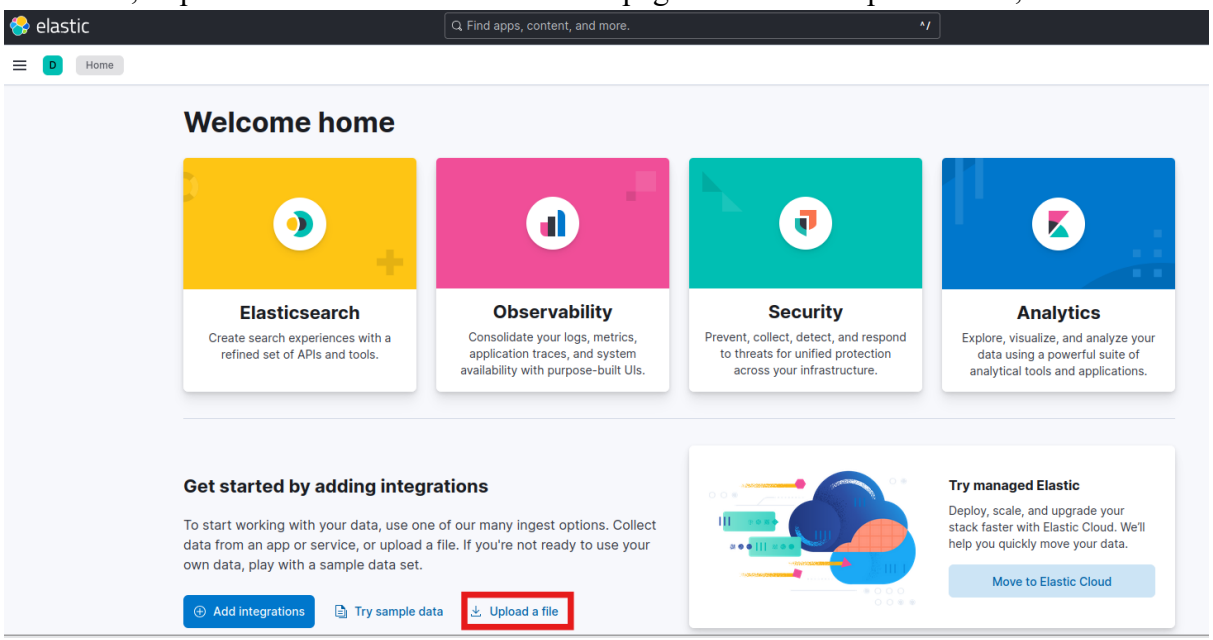
First, I created a table in google sheets. Below image is the table.

Table1

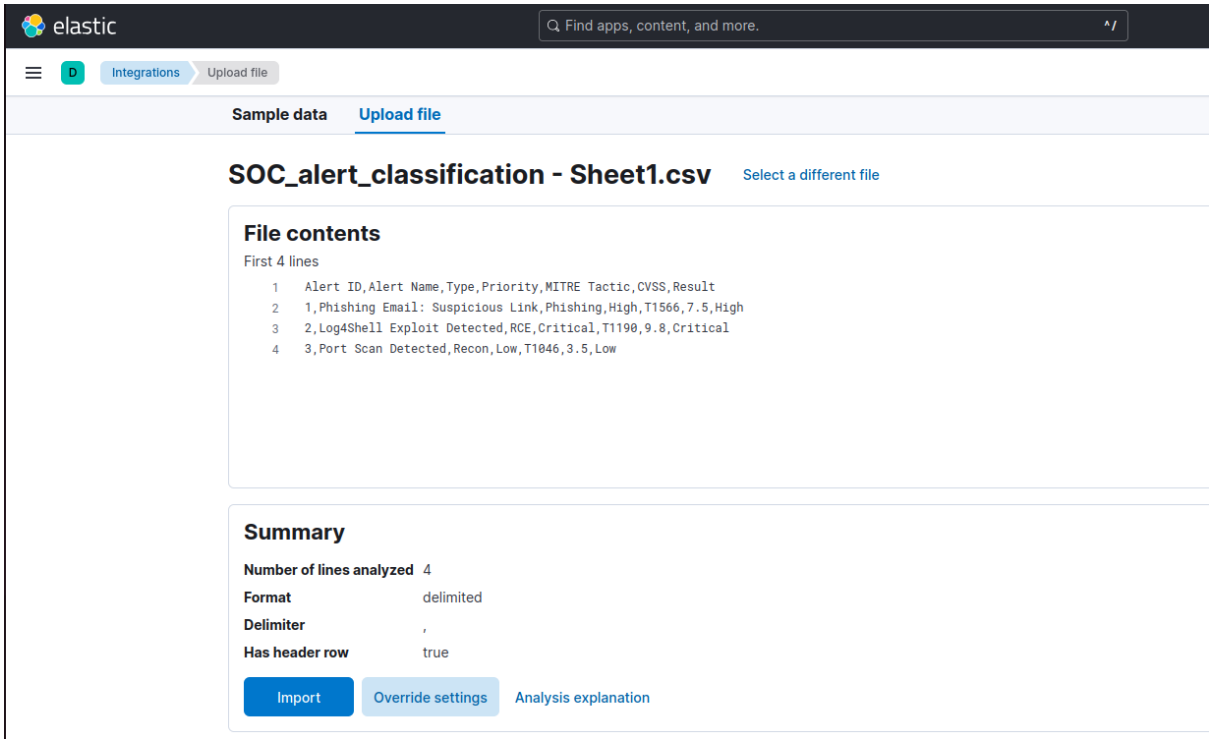
Alert ID	Alert Name	Type	Priority	MITRE Tactic	CVSS	Result
1	Phishing Email: Suspicious Link	Phishing	High	T1566	7.5	High
2	Log4Shell Exploit Detected	RCE	Critical	T1190	9.8	Critical
3	Port Scan Detected	Recon	Low	T1046	3.5	Low

and then I saved and downloaded it as a CSV file.

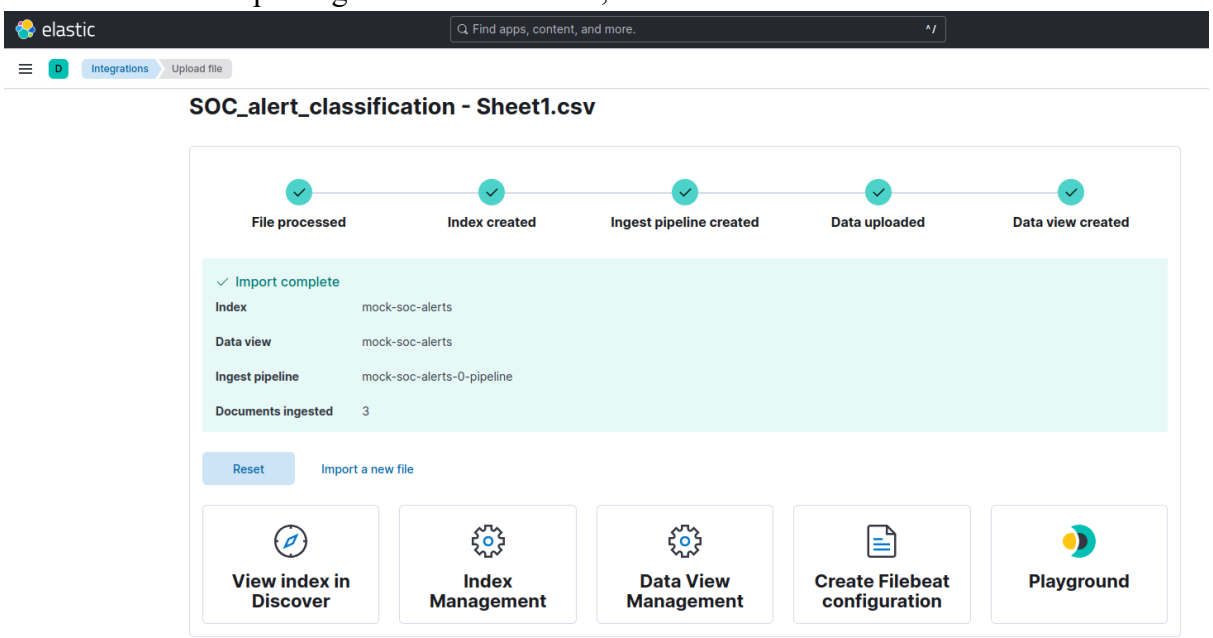
After that, I opened kibana and in kibana home page I clicked on upload a file,



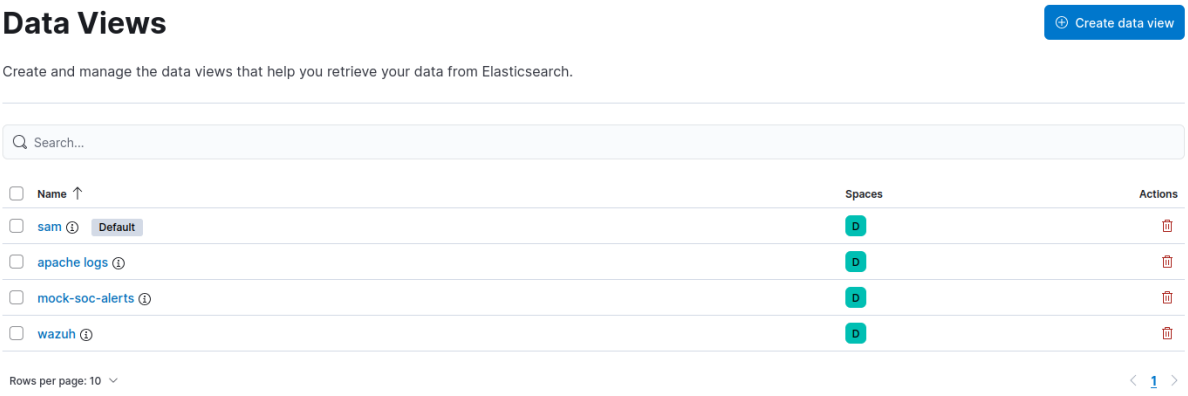
After clicking on it, it opened a webpage and the screenshot of that webpage is,



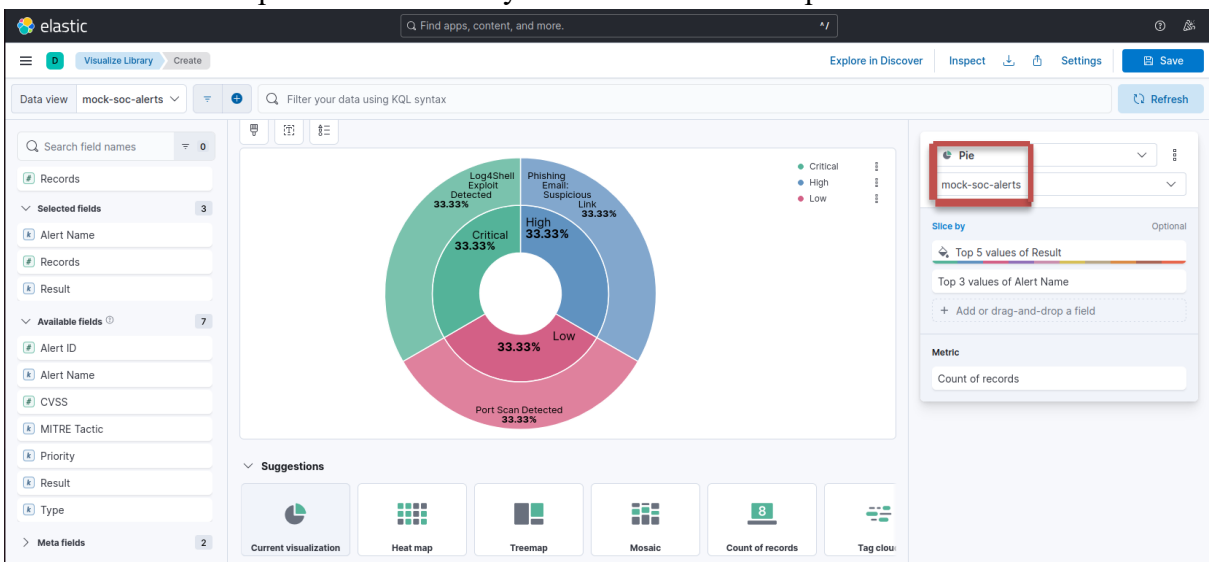
I clicked on import, gave a name as mock-sock-alerts and again clicked on import. The screenshot after importing the file is as below,



After importing that CSV file, I opened data views and, in that page, we can see mock-soc-alerts. The screenshot of that page is below,



Now to get a pie chart of it, go to visualize library, then click on lens.
After clicking on it, I dragged the Records field in the screen, select Pie in the right side of the page then write mock-sock-alerts in index which is below pie. I dragged the result field and alert name and put those in slice by. And the result is the pie chart in below screenshot.



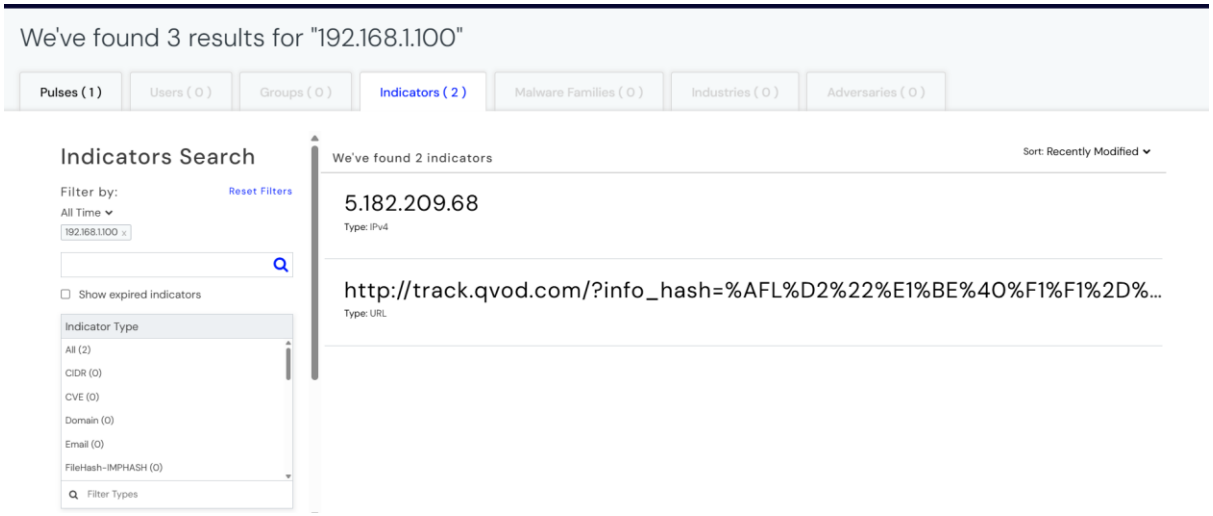
Draft a 100-word email to escalate a Critical alert to Tier 2, summarizing the incident and IOCs.

Dear Tier 2 Team,
A Critical alert has been triggered involving multiple failed authentication attempts against the Administrator account on host WIN-11-01 (192.168.1.25). Wazuh detected 20 failed logon events (Event ID 4625) within a 3-minute window, indicating a potential brute-force attack. The activity originated from source IP 192.168.1.50. No successful login has been observed so far. The source IP has been temporarily blocked at the firewall, and the affected account password has been reset as a precaution.
Please review for deeper log analysis, persistence checks, and potential lateral movement.
Regards,
Sai Charan
Tier 1 SOC Analyst

Alert Triage Practice

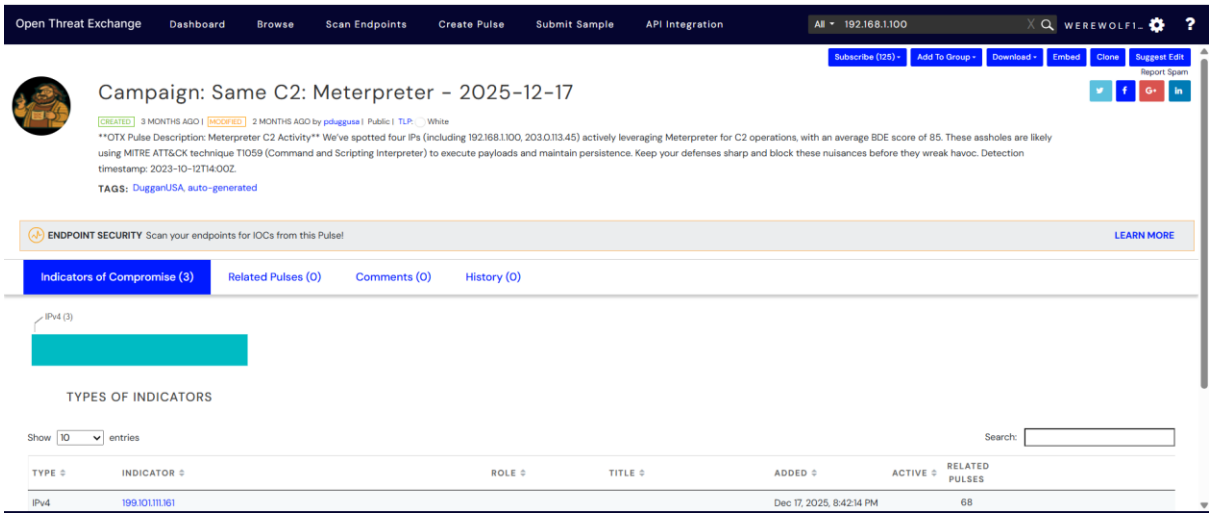
Alert ID	Description	Source IP	Priority	Status
002	Brute force - SSH	192.168.1.100	Medium	Open

In the search bar in AlienVault OTX, I entered the IP address 192.168.1.100, and the output is the screenshot below.

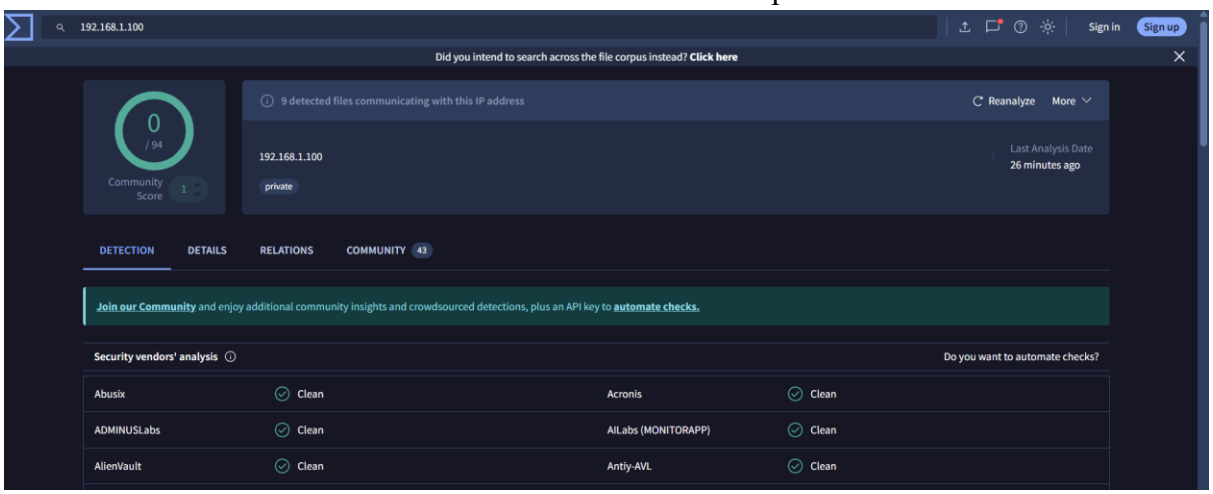


When I clicked on pulses, the output is the screenshot below,

Now after clicking on the campaign: Same C2: Meterpreter, I got the output of screenshot below.



I also entered the same IP address in virus total and the output is the screenshot below.

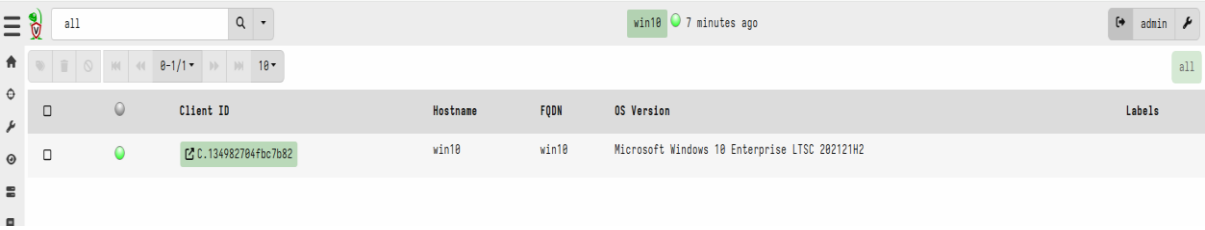


Findings

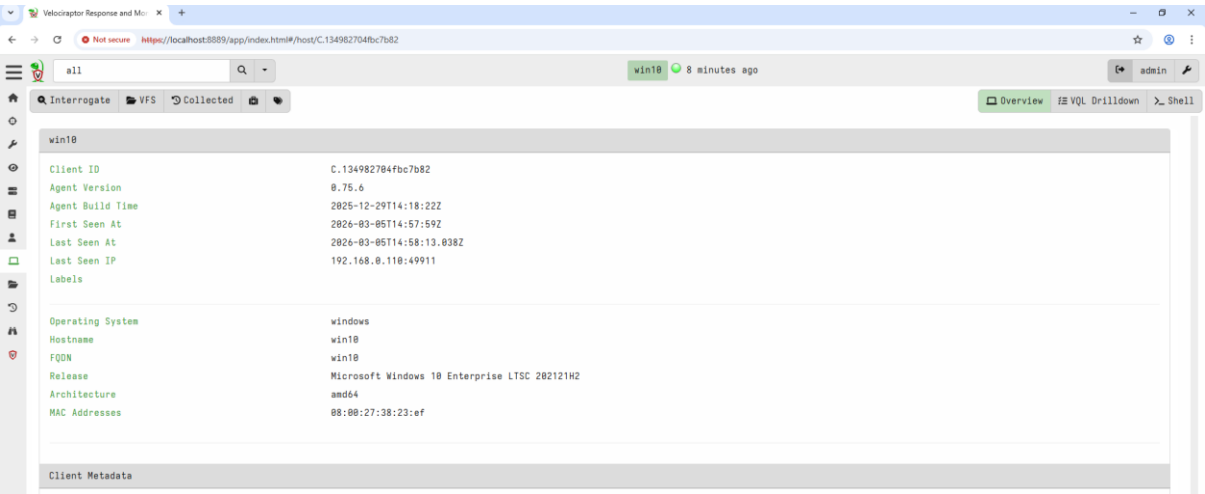
The source IP was analysed using AlienVault OTX and VirusTotal threat intelligence platforms. The IP address showed limited malicious history and was not widely associated with known threat campaigns. The activity may represent a scanning attempt or misconfigured internal system rather than a confirmed attack. Continuous monitoring and log analysis were recommended.

Use Velociraptor to collect network connections

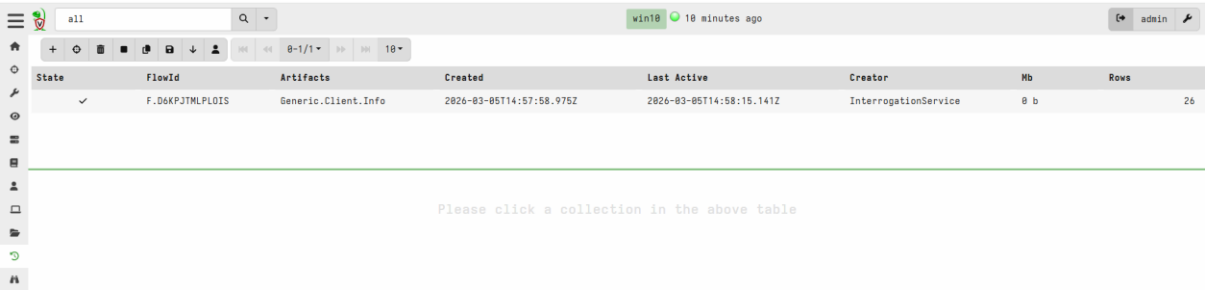
I installed 2 windows VM where one is server, another one is client using velociraptor.
In server side I opened the dashboard of velociraptor and clicked on search symbol and the output is the screenshot below



Then I clicked on client ID and it the screenshot of it is,

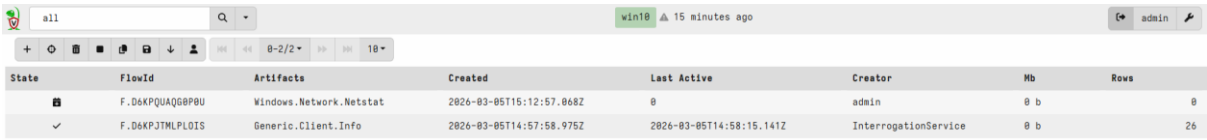


Then I clicked on collected which is on upper left side. The screenshot of it is below.

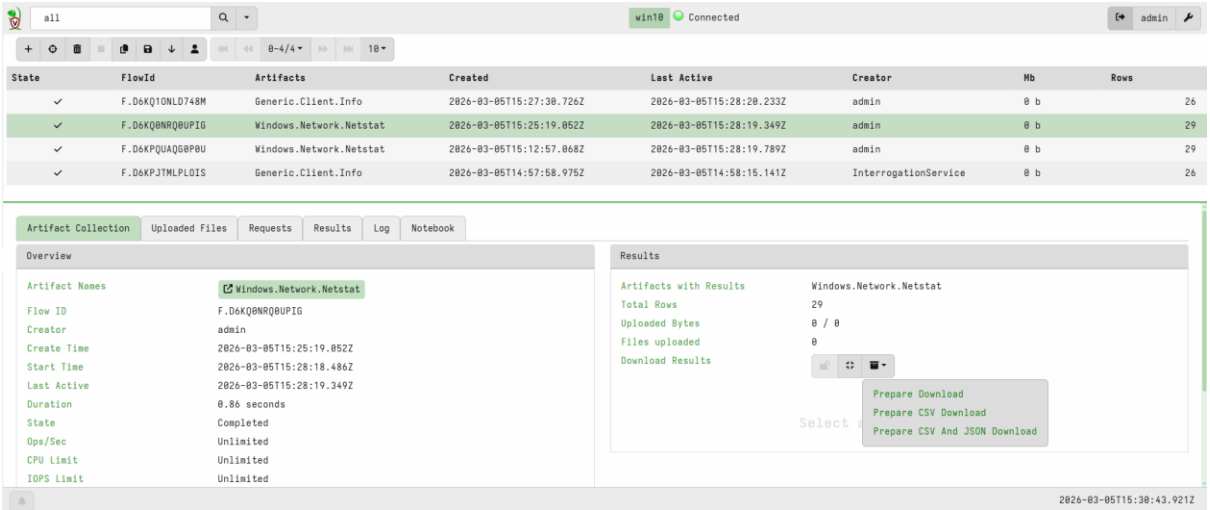


Then I clicked on + sign on top left side then in the search bar I typed netstat and selected windows.network.netstat and output of that is below,

After that I clicked on Launch. The output screenshot of it is,



Now I clicked on the new field where artifact is windows.network.netstat after the state is in tick mark and then downloaded CSV file. The screenshot of it is below.



The screenshot of result tab is

State	FlowId	Artifacts	Created	Last Active	Creator	Mb	Rows
✓	F.D6KQ1ONLD748M	Generic.Client.Info	2026-03-05T15:27:30.726Z	2026-03-05T15:28:20.233Z	admin	0 b	26
✓	F.D6KQ0NRQ0UPIG	Windows.Network.Netstat	2026-03-05T15:25:19.052Z	2026-03-05T15:28:19.349Z	admin	0 b	29
✓	F.D6KPQUAQG0P0U	Windows.Network.Netstat	2026-03-05T15:12:57.068Z	2026-03-05T15:28:19.789Z	admin	0 b	29
✓	F.D6KPJTMLPLOIS	Generic.Client.Info	2026-03-05T14:57:58.975Z	2026-03-05T14:58:15.141Z	InterrogationService	0 b	26

Artifact Collection | Uploaded Files | Requests | Results | Log | Notebook

Windows.Network.Netstat

0-10/29 10

Pid	Name	Family	Type	Status	Laddr.IP	Laddr.Port	Raddr.IP	Raddr.Port	Timestamp
884	svchost.exe	IPv4	TCP	LISTEN	0.0.0.0	135	0.0.0.0	0	2026-03-05T14:04:52Z
4	System	IPv4	TCP	LISTEN	192.168.0.110	139	0.0.0.0	0	2026-03-05T14:04:57Z
4572	svchost.exe	IPv4	TCP	LISTEN	0.0.0.0	5040	0.0.0.0	0	2026-03-05T14:06:46Z
660	lsass.exe	IPv4	TCP	LISTEN	0.0.0.0	49664	0.0.0.0	0	2026-03-05T14:04:52Z
540	wininit.exe	IPv4	TCP	LISTEN	0.0.0.0	49665	0.0.0.0	0	2026-03-05T14:04:53Z
1876	spoolsv.exe	IPv4	TCP	LISTEN	0.0.0.0	49666	0.0.0.0	0	2026-03-05T14:04:57Z
1036	svchost.exe	IPv4	TCP	LISTEN	0.0.0.0	49667	0.0.0.0	0	2026-03-05T14:04:58Z

Collect a memory dump and hash it using sha256sum

To do this, again click on plus (+) sign in velociraptor and type windows.memory and select windows.memory.acquisition and click on launch.

New Collection: Select Artifacts to collect

windows.memory

- Windows.Memory.Acquisition
- Windows.Memory.Intezer
- Windows.Memory.PEDump
- Windows.Memory.ProcessDump
- Windows.Memory.ProcessInfo
- Windows.System.VAD

Windows.Memory.Acquisition

Type: client

Acquires a full memory image by using the built-in WinPmem driver.

NOTE: This artifact usually transfers a lot of data. You should increase the default timeout to allow it to complete.

Memory images are typically susceptible to a lot of smear. To minimize this we need to acquire memory as quickly as possible. This artifact offers a few compression methods for the output file. Reducing the size of the file will decrease time needed for IO but will increase CPU requirements so this is a trade-off. Empirically we found that using S2 compression gives a reasonable compression and very high speed reducing acquisition time from the no compression options significantly.

To decompress the image you can use the Go WinPmem binary

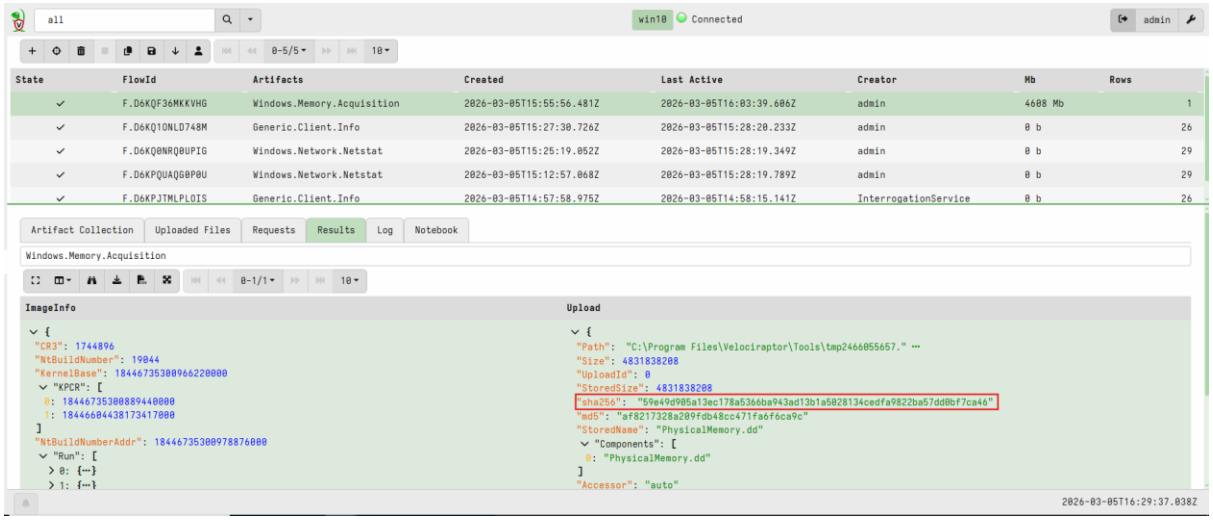
```
go-winpmem.exe expand image.compressed image.raw
```

Parameters

Name	Type	Default	Description
ServiceName			Override the name of the driver service to install.

Select Artifacts | Configure Parameters | Specify Resources | Review | Launch

After the process is complete, to check the SHA256 go to results and there will be sha256in there, the screenshot of that is,



Item	Description	Collected by	Date	Hash Value (SHA 256)
Memory dump	Windows 10 dump	SOC analyst	05-03-2026	59e49d905a13ec178a5366ba943ad13b1a5028134cedfa9822ba57dd0bf7ca46

Simulate an attack, detect, triage, respond, and document.

To simulate an attack, I opened 3 virtual machines, one is kali linux(attacker), second one is vulnerable ubuntu machine(target), third one is ubuntu machine which has ELK stack and wazuh manager.
Target machine has wazuh agent and sends its logs to the wazuh manager.
Now I opened kali linux machine and used nmap for host discovery, then used Metasploit for sending malicious payload to access the system. I used ProFTPD 1.3.3c backdoor command execution for the exploit.
After selecting payloads and configuring options, I typed the command below in Metasploit which is,

- exploit

After that I go to ubuntu virtual machine which has wazuh manager and typed the command below

- sudo tail -f /var/ossec/logs/alerts/alerts.log

and the logs show that some IP address is connected to target machine using FTP connection. The output of those logs is taken as screenshot and the screenshot of those logs are below.

```
** Alert 1772783827.683713: - syslog,proftpd,connection_attempt,gdpr_IV_32.2,hipaa_164.312.b,nist_800_53_AC.7,
nist_800_53_AU.14,pci_dss_10.2.5,tsc_CC6.8,tsc_CC7.2,tsc_CC7.3,
2026 Mar 06 13:27:07 (vtcsec) any->journald
Rule: 11201 (level 3) -> 'ProFTPD: FTP session opened.'
Src IP: 192.168.0.104
Mar 06 07:57:06 vtcsec proftpd[3595]: vtcsec (192.168.0.104[192.168.0.104]) - FTP session opened.

** Alert 1772783827.684108: - syslog,proftpd,connection_attempt,gdpr_IV_32.2,hipaa_164.312.b,nist_800_53_AC.7,
nist_800_53_AU.14,pci_dss_10.2.5,tsc_CC6.8,tsc_CC7.2,tsc_CC7.3,
2026 Mar 06 13:27:07 (vtcsec) any->/var/log/syslog
Rule: 11201 (level 3) -> 'ProFTPD: FTP session opened.'
Src IP: 192.168.0.104
Mar  6 02:57:06 vtcsec proftpd[3595]: vtcsec (192.168.0.104[192.168.0.104]) - FTP session opened.

root@sai-VMware-Virtual-Platform:/home/sai#
```

In the above screenshot you can see the time of the attack, source IP address which is attacker IP address.

In kibana dashboard, the logs are

Timestamp	Source IP	Alert description	MITRE technique
2026-03-06 13:27:07	192.168.0.104	ProFTPD exploit	T1210

Now to block the IP address through crowdsec, I first installed crowdsec with the commands below, they are,

- curl -s https://install.crowdsec.net | sudo sh
- sudo apt update
- sudo apt install crowdsec
- sudo apt install crowdsec-firewall-bouncer-iptables
- sudo apt install crowdsec-firewall-bouncer-nftables

Now to block the attacker IP address I typed the command below, which is,

- sudo cscli decisions add --ip 192.168.0.104 –duration 1h

```
marlinspike@vtcsec:~$ sudo cscli decisions add --ip 192.168.0.104 --duration 1h
INFO Decision successfully added
```

Now when I ping the IP address the output is the screenshot below.

```
marlinspike@vtcsec:~$ ping 192.168.0.104
PING 192.168.0.104 (192.168.0.104) 56(84) bytes of data.
```

Stakeholder Briefing – FTP Exploit Incident

A security alert was detected in the SOC lab environment involving an attempted unauthorized access to a server running an outdated FTP service. The attack originated from an internal test machine using a known vulnerability in the ProFTPD service. Our monitoring system, Wazuh, successfully detected the suspicious activity and generated alerts for investigation. The security team analyzed the logs and confirmed the exploit attempt. As an immediate response, the attacker's IP address was blocked using CrowdSec to prevent further communication with the server. No data loss or system compromise occurred. The incident was contained quickly, and recommendations have been made to update vulnerable services and strengthen monitoring controls.